Deduction theorem for the quantifier-rule calculus

The Open Logic Project contributors

openlogic:first-order-logic/axiomatic-deduction:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-first-order-logic-axiomatic-deduction/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: pairing sibling of thm:deduction-thm (this batch); the basis and MP cases cite it by name, the QR case is carried in-span; in-source stub thm:deduction-thm.

Editorial note. The label thm:deduction-thm-q names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own thm environment in the unit source.

Allowed standard prerequisites (editorial).

- Propositional deduction theorem. Same unit, \ollabel{thm:deduction-thm} (source line 643, this batch). The induction basis and the MP case are 'identical to' / 'as in' its proof, delivered as a sibling card.
- Quantifier rule bookkeeping. Same unit, QR rule with eigenvariable condition. The inductive step for the universal introduction rule uses the eigenvariable condition on a.

Source reference. Original theorem thm-deduction-thm: sources/olp-first-order-logic-axiomatic-deduction/pinned-original-source.tex, source lines 643–643. Prerequisite pointer invoked at proof line(s) 559; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Theorem 0.1 (Deduction Theorem)**.** *If* $\Gamma \cup \{A\} \vdash B$*, then* $\Gamma \vdash A \to B$*.*

Proof. We again proceed by induction on the length of the derivation of B from $\Gamma \cup \{A\}$.

The proof of the induction basis is identical to that in the proof of thm-deduction-thm.

For the inductive step, suppose again that the derivation of B from $\Gamma \cup \{A\}$ ends with a step B which is justified by an inference rule. If the inference rule is modus ponens, we proceed as in the proof of thm-deduction-thm. If the

inference rule is QR, we know that $B \equiv C \to \forall x\, D(x)$ and formula of the form $C \to D(a)$ appears earlier in the derivation, where a does not occur in C, A, or Γ. We thus have that

$$\Gamma \cup \{A\} \vdash C \to D(a),$$

and the induction hypothesis applies, i.e., we have that

$$\Gamma \vdash A \to (C \to D(a)).$$

By

$$\vdash (A \to (C \to D(a))) \to ((A \wedge C) \to D(a))$$

and modus ponens we get

$$\Gamma \vdash (A \wedge C) \to D(a).$$

Since the eigenvariable condition still applies, we can add a step to this derivation justified by QR, and get

$$\Gamma \vdash (A \wedge C) \to \forall x\, D(x).$$

We also have

$$\vdash ((A \wedge C) \to \forall x\, D(x)) \to (A \to (C \to \forall x\, D(x)),$$

so by modus ponens,

$$\Gamma \vdash A \to (C \to \forall x\, D(x)),$$

i.e., $\Gamma \vdash A \to B$.

We leave the case where B is justified by the rule QR, but is of the form $\exists x\, D(x) \to C$, as an exercise. □